

Assignment 4 Write Up

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Dataset I = Amazon Stock Dataset

Dataset II = Iris Dataset

Question 1.

Tools and Libraries:

- Python - make script
- pandas - read CSVs and handle dataframes
- pyyaml to convert the yaml metadata file of dataset I to a python dictionary
- ucimlrepo to import the iris dataset
- h5py for HDF5 input and output
- json to convert python objects into json objects

Procedure:

1. Read in the information (pd.read_csv for dataset II and ucimlrepo.fetch_ucirepo(id=53) for dataset I)
2. Create a combined dataframe (features + targets for dataset II and direct dataframe for dataset I)
3. Clean metadata (convert YAML objects into json safe types for dataset I)
4. Open the HDF5 file in write mode
5. Add metadata to HDF5 file as attributes
6. Create data groupings and store each dataframe column as a different dataset
7. Added dataset attributes to HDF5 file

Output

- [Conversion Code](#)
- [Dataset I HDF5 file output](#)
- [Dataset II HDF5 file output](#)

Question 2a.

- *Logical Organization:* Originally organized data in a flat tabular representation with metadata in a separate YAML file or as an object. The logical model is a single table with column names and a separate container for metadata. The HDF5 result contains a hierarchical model, where top level attributes store dataset level metadata. One group contains the others through per column datasets. This mapping preserves the logical table but also allows it to be expressed as column oriented datasets in the group.
- *Physical Organization:* The original format for both datasets was CSV files that were plain text, row major, single file, and had no embedded metadata. This was inefficient for random column reading and partial loading. The new format for both datasets is HDF5, which contains binary, typed datasets with the metadata embedded in the dataset attributes. Data is stored in column datasets which allows for faster random column access and better input and output for large datasets.

- *Metadata embedded vs external:* The core descriptive and technical metadata is embedded as file level attributes in the HDF5 format. The HDF5 file names for dataset I and dataset II are “AMZN.h5” and “iris.h5” respectively.

Question 2b.

Additional Metadata and/or information I would include for cataloguing and preservation purposes:

- Change Log/Version number (add to both datasets)
- License (add to dataset II, dataset I contains it already)
- Data collection methodology (add to dataset I, dataset II contains it already)
- Number of features (add to dataset I, dataset II contains it already)
- Number of instances (add to dataset I, dataset II contains it already)
- Target column (add to dataset I, dataset II contains it already)
- Task (add to dataset I, dataset II contains it already)
- Feature Types (add to dataset I, dataset II contains it already)
- Abstract (add to dataset I, dataset II contains it already)